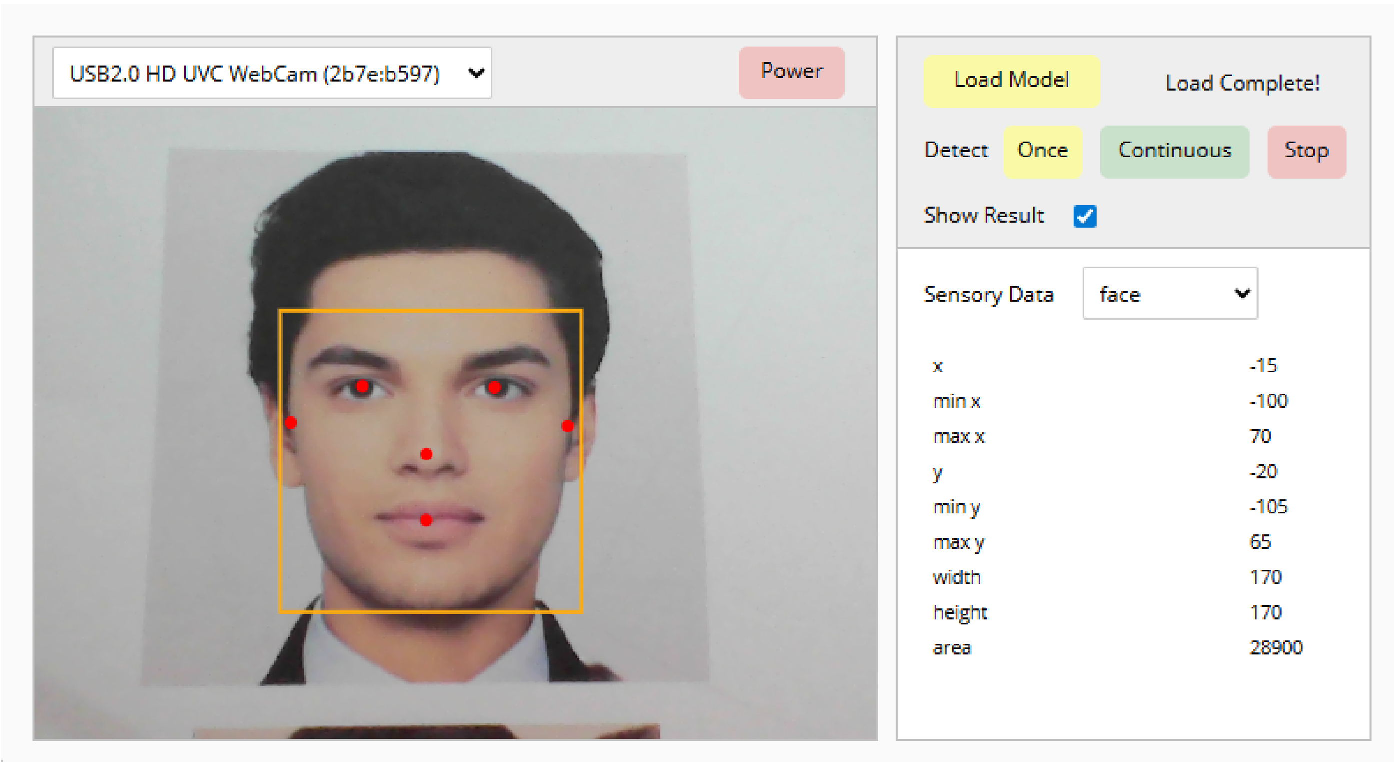
Open Dashboard

Open Dashboard is not a block that can be used in block coding,
but it allows you to open a dashboard where you can check how the model used in the extension module is applied.

Dashboard Screen

When you click Open Dashboard, you can see the following screen.



Detailed Features

Power

Turns the selected camera on or off.

Load Model

Loads the trained face model. This is a required step to use the **Face Detection** extension module.

Detect

Starts or stops face detection.
You can choose whether to run it only once with the Once button, or continuously with the Continuous

button.

You can also stop detection using the Stop button.

Show Result

Displays face detection results on the camera screen.

Sensory Data

Displays the face data values found by face detection.

You can check the x, y coordinates of the selected face part.

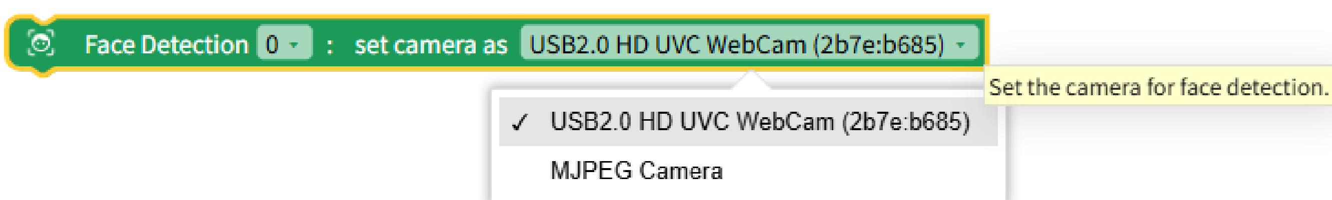
Dropdown Options and Input Values

Name	Type	Description	Range / Type
part	Dropdown Option	Face part	face, ear, eye, nose, mouth

Blocks

Select Camera

Selects the camera to use for the Face Detection module.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
camera	Dropdown Option	Camera to use	Connected camera list

JavaScript Code

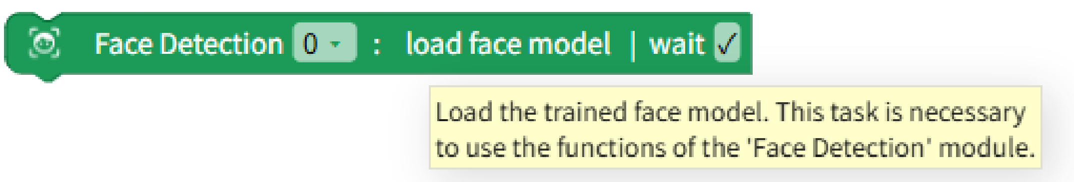
```
// Set a specific camera for face detection (id is an example)
$('FaceDetection*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb';
```

Python Code

```
# Set a specific camera for face detection (id is an example)
__('FaceDetection*0:camera.deviceId').d =
'035658da47183882a695a82c45b8f3e9ae50cef47945ccdc3f31e1ae1fbca9cb'
```

Load Face Model

Loads the trained face model. This must be done to use the features of the Face Detection module.
If **Wait** is checked, it waits until the model finishes loading.
If **Wait** is checked, it can only be used inside an async function.



JavaScript Code

```
// Load face model | Wait: O
$('FaceDetection*0:load_model').d = 1;
await $('FaceDetection*0:!load_model').w();

// Load face model | Wait: X
$('FaceDetection*0:load_model').d = 1;
```

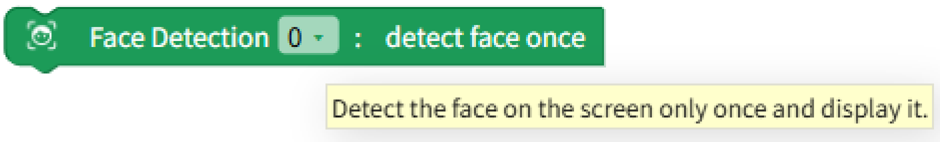
Python Code

```
# Load face model | Wait: O
__('FaceDetection*0:load_model').d = 1
```

```
await __('FaceDetection*0:!load_model').w()

# Load face model | Wait: X
__('FaceDetection*0:load_model').d = 1
```

Detect Face Once

Detects faces currently on the screen and displays them only once.



JavaScript Code

```
// Detect face once
$('FaceDetection*0:detect.once').d = 1;
```

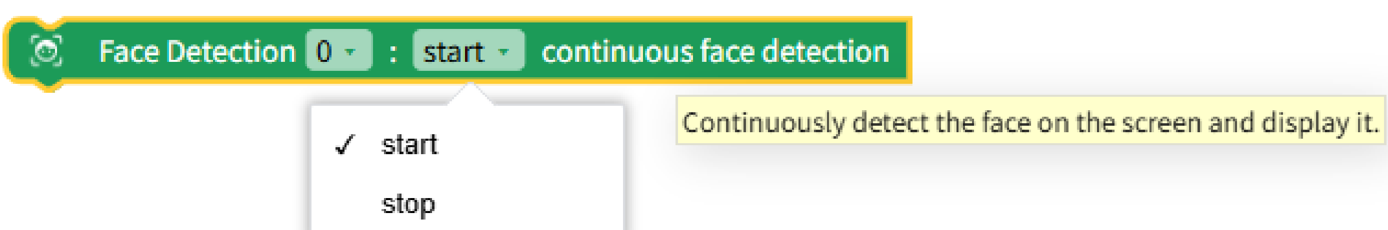
Python Code

```
# Detect face once
__('FaceDetection*0:detect.once').d = 1
```

Detect Face Continuously

Starts or stops continuous face detection.

When continuous detection starts, it keeps tracking and displaying faces currently on the screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown Option	Face detection	Start (1), Stop (0)

JavaScript Code

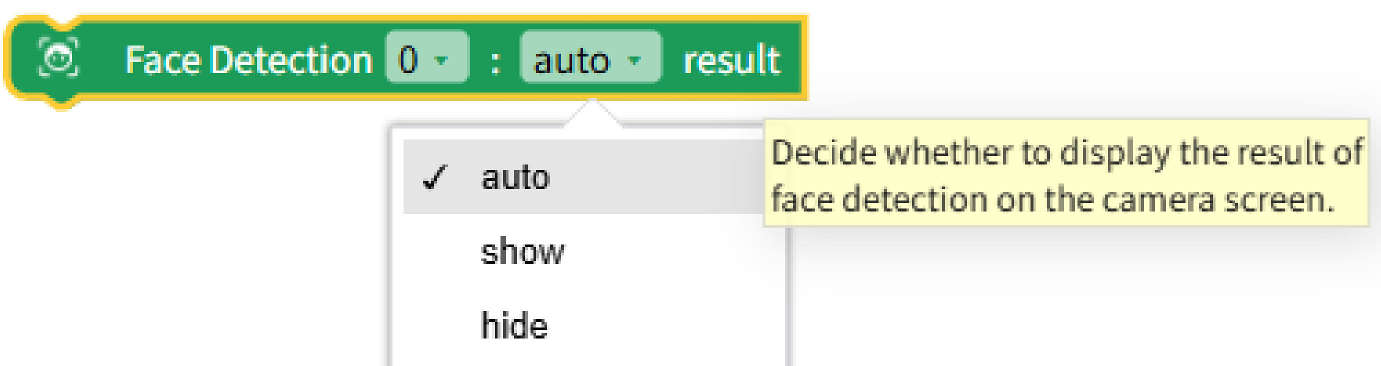
```
// Start continuous face detection
$('FaceDetection*0:detect.continuous').d = 1;

// Stop continuous face detection
$('FaceDetection*0:detect.continuous').d = 0;
```

Python Code

```
# Start continuous face detection
__('FaceDetection*0:detect.continuous').d = 1

# Stop continuous face detection
__('FaceDetection*0:detect.continuous').d = 0
```

Show Face Detection Result

Decides whether to display face detection results on the camera screen.



Dropdown Options and Input Values

Name	Type	Description	Range / Type
toggle	Dropdown Option	Show results	Show (1), Hide (0)

JavaScript Code

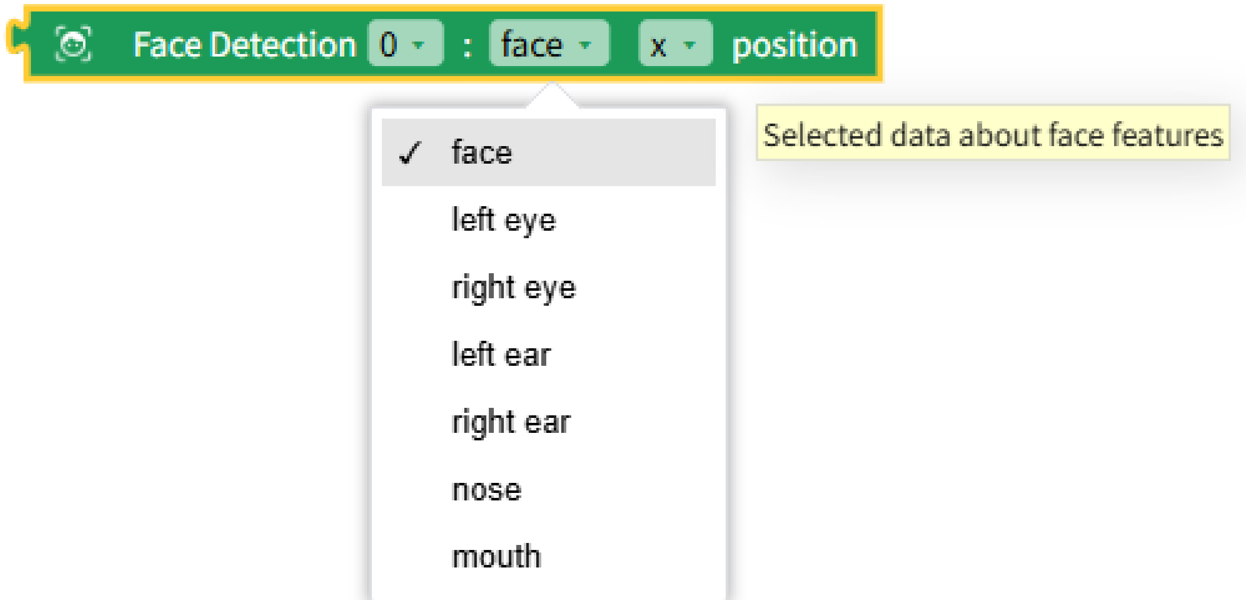
```
// Show face detection results
$('FaceDetection*0:display').d = 1;

// Hide face detection results
$('FaceDetection*0:display').d = 0;
```

Python Code

```
# Show face detection results
__('FaceDetection*0:display').d = 1

# Hide face detection results
__('FaceDetection*0:display').d = 0
```

Face Data

Returns the face data obtained from face detection.



Name	Type	Description	Range / Type
part	Dropdown Option	Face part	face, eye.left, eye.right, ear.left, ear.right, nose, mouth
axis	Dropdown Option	Coordinate axis	x, y

JavaScript Code

```
// Face x coordinate
$('FaceDetection*0:face.x').d;

// Face y coordinate
$('FaceDetection*0:face.y').d;

// Left eye x coordinate
$('FaceDetection*0:eye.left.x').d;

// Left eye y coordinate
$('FaceDetection*0:eye.left.y').d;

// Right eye x coordinate
```

```
$('FaceDetection*0:eye.right.x').d;

// Right eye y coordinate
$('FaceDetection*0:eye.right.y').d;

// Left ear x coordinate
$('FaceDetection*0:ear.left.x').d;

// Left ear y coordinate
$('FaceDetection*0:ear.left.y').d;

// Right ear x coordinate
$('FaceDetection*0:ear.right.x').d;

// Right ear y coordinate
$('FaceDetection*0:ear.right.y').d;

// Nose x coordinate
$('FaceDetection*0:nose.x').d;

// Nose y coordinate
$('FaceDetection*0:nose.y').d;

// Mouth x coordinate
$('FaceDetection*0:mouth.x').d;

// Mouth y coordinate
$('FaceDetection*0:mouth.y').d;
```

Python Code

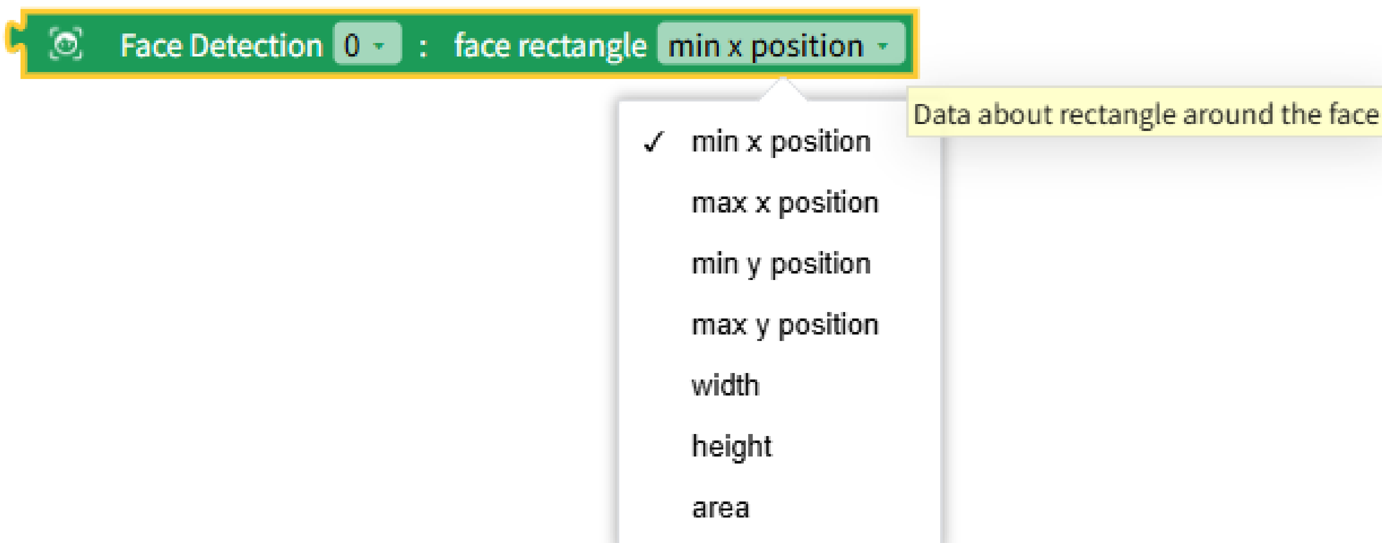
```
# Face x coordinate
__('FaceDetection*0:face.x').d
```

```
# Face y coordinate
__('FaceDetection*0:face.y').d

# Left eye x coordinate
__('FaceDetection*0:eye.left.x').d

# Left eye y coordinate
__('FaceDetection*0:eye.left.y').d

# Right eye x coordinate
__('FaceDetection*0:eye.right.x').d

# Right eye y coordinate
__('FaceDetection*0:eye.right.y').d

# Left ear x coordinate
__('FaceDetection*0:ear.left.x').d

# Left ear y coordinate
__('FaceDetection*0:ear.left.y').d

# Right ear x coordinate
__('FaceDetection*0:ear.right.x').d

# Right ear y coordinate
__('FaceDetection*0:ear.right.y').d

# Nose x coordinate
__('FaceDetection*0:nose.x').d

# Nose y coordinate
__('FaceDetection*0:nose.y').d

# Mouth x coordinate
```

```
__('FaceDetection*0:mouth.x').d

# Mouth y coordinate
__('FaceDetection*0:mouth.y').d
```

Face Bounding Box Data

Defines the area around the detected face as a rectangle and returns that rectangle's data.



Name	Type	Description	Range / Type
data	Dropdown Option	Rectangle data	min_x, max_x, min_y, max_y, width, height, area

JavaScript Code

```
// Face rectangle min x
$('FaceDetection*0:face.min_x').d;

// Face rectangle max x
$('FaceDetection*0:face.max_x').d;

// Face rectangle min y
```

```
$('FaceDetection*0:face.min_y').d;

// Face rectangle max y
$('FaceDetection*0:face.max_y').d;

// Face rectangle width
$('FaceDetection*0:face.width').d;

// Face rectangle height
$('FaceDetection*0:face.height').d;

// Face rectangle area
$('FaceDetection*0:face.area').d;
```

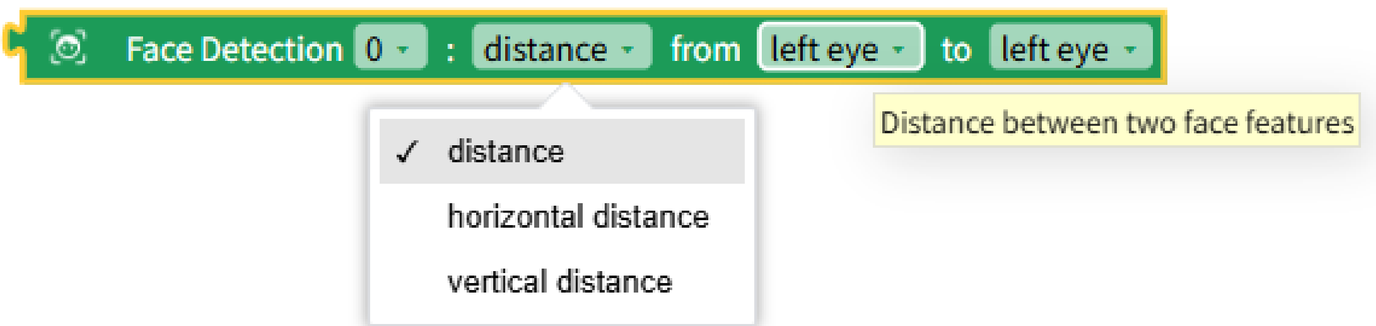
Python Code

```
# Face rectangle min x
__('FaceDetection*0:face.min_x').d

# Face rectangle max x
__('FaceDetection*0:face.max_x').d

# Face rectangle min y
__('FaceDetection*0:face.min_y').d

# Face rectangle max y
__('FaceDetection*0:face.max_y').d

# Face rectangle width
__('FaceDetection*0:face.width').d

# Face rectangle height
__('FaceDetection*0:face.height').d
```

```
# Face rectangle area
__('FaceDetection*0:face.area').d
```

Distance Between Face Parts

Returns the distance between two face parts using face detection data.



Name	Type	Description	Range / Type
part1	Dropdown Option	Face part	face, eye.left, eye.right, ear.left, ear.right, nose, mouth
part2	Dropdown Option	Face part	face, eye.left, eye.right, ear.left, ear.right, nose, mouth
distance	Dropdown Option	Distance type	distance, horizontal distance, vertical distance

JavaScript Code

```
// Distance from left eye to left eye
Math.sqrt( Math.pow(($('FaceDetection*0:eye.left.x').d -
$('FaceDetection*0:eye.left.x').d), 2) +
Math.pow(($('FaceDetection*0:eye.left.y').d -
$('FaceDetection*0:eye.left.y').d), 2) );

// Horizontal distance from left eye to right eye
```

```
Math.abs($('FaceDetection*0:eye.right.x').d -
$('FaceDetection*0:eye.left.x').d);

// Vertical distance from left eye to left ear
Math.abs($('FaceDetection*0:ear.left.y').d - $('FaceDetection*0:eye.left.y').d);

// Distance from left eye to right ear
Math.sqrt( Math.pow(($('FaceDetection*0:ear.right.x').d -
$('FaceDetection*0:eye.left.x').d), 2) +
Math.pow(($('FaceDetection*0:ear.right.y').d -
$('FaceDetection*0:eye.left.y').d), 2) );

// Distance from left eye to nose
Math.sqrt( Math.pow(($('FaceDetection*0:nose.x').d -
$('FaceDetection*0:eye.left.x').d), 2) + Math.pow(($('FaceDetection*0:nose.y').d
- $('FaceDetection*0:eye.left.y').d), 2) );

// Distance from left eye to mouth
Math.sqrt( Math.pow(($('FaceDetection*0:mouth.x').d -
$('FaceDetection*0:eye.left.x').d), 2) + Math.pow(($('FaceDetection*0:mouth.y').d
- $('FaceDetection*0:eye.left.y').d), 2) );
```

Python Code

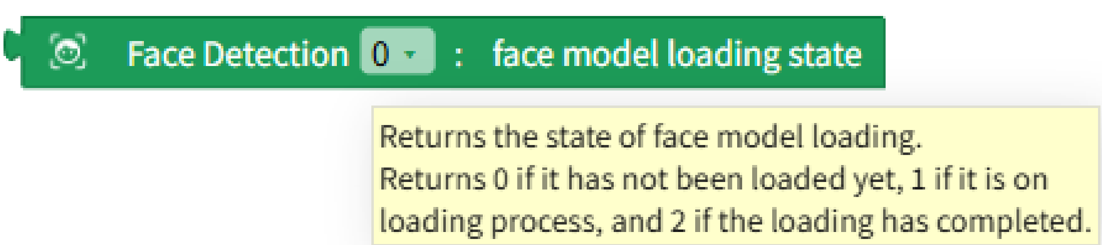
```
# Distance from left eye to left eye
math.sqrt( math.pow((__('FaceDetection*0:eye.left.x').d -
__('FaceDetection*0:eye.left.x').d), 2) +
math.pow((__('FaceDetection*0:eye.left.y').d -
__('FaceDetection*0:eye.left.y').d), 2) )

# Horizontal distance from left eye to right eye
math.fabs(__('FaceDetection*0:eye.right.x').d -
__('FaceDetection*0:eye.left.x').d)
```

```
# Vertical distance from left eye to left ear
math.fabs(__('FaceDetection*0:ear.left.y').d -
__('FaceDetection*0:eye.left.y').d)

# Distance from left eye to right ear
math.sqrt( math.pow((__('FaceDetection*0:ear.right.x').d -
__('FaceDetection*0:eye.left.x').d), 2) +
math.pow((__('FaceDetection*0:ear.right.y').d -
__('FaceDetection*0:eye.left.y').d), 2) )

# Distance from left eye to nose
math.sqrt( math.pow((__('FaceDetection*0:nose.x').d -
__('FaceDetection*0:eye.left.x').d), 2) +
math.pow((__('FaceDetection*0:nose.y').d - __('FaceDetection*0:eye.left.y').d),
2) )

# Distance from left eye to mouth
math.sqrt( math.pow((__('FaceDetection*0:mouth.x').d -
__('FaceDetection*0:eye.left.x').d), 2) +
math.pow((__('FaceDetection*0:mouth.y').d - __('FaceDetection*0:eye.left.y').d),
2) )
```

Face Model State Value

Returns the face model loading state.

Returns 0 if not loaded, 1 if loading, and 2 if loading is complete.



JavaScript Code

```
// Face model state value
$('FaceDetection*0:model_state').d;
```

Python Code

```
# Face model state value
__('FaceDetection*0:model_state').d
```

Was a Face Detected?

Returns whether a face was detected as **True (1) / False (0)**.

JavaScript Code

```
// Was a face detected?
$('FaceDetection*0:detected').d;
```

Python Code

```
# Was a face detected?
__('FaceDetection*0:detected').d
```